```
Last login: Tue Jun 16 12:21:13 on ttys002
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/HPC-Upload\ 2/mpi-patterns
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % ls
adjacent_communication.png      adjacent_trace.csv              mpi_adjacent.c                  mpi_mirrored.c
adjacent_latency.png            mpi_adjacent                    mpi_halfshift.c                 visualize_trace.py
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_adjacent.c -o mpi_adjacent
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f adjacent_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 2 ./mpi_adjacent 100000
Rank 0: Sending 100000 elements to Rank 1
Rank 1: Received 100000 elements from Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat adjacent_trace.csv
0,1,400000,0.000155
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f adjacent_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 4 ./mpi_adjacent 100000
Rank 2: Sending 100000 elements to Rank 3
Rank 0: Sending 100000 elements to Rank 1
Rank 3: Received 100000 elements from Rank 2 (First element: 20)
Rank 1: Received 100000 elements from Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat adjacent_trace.csv
0,1,400000,0.000664
2,3,400000,0.000662
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f adjacent_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 100000
Rank 6: Sending 100000 elements to Rank 7
Rank 2: Sending 100000 elements to Rank 3
Rank 4: Sending 100000 elements to Rank 5
Rank 0: Sending 100000 elements to Rank 1
Rank 3: Received 100000 elements from Rank 2 (First element: 20)
Rank 1: Received 100000 elements from Rank 0 (First element: 0)
Rank 5: Received 100000 elements from Rank 4 (First element: 40)
Rank 7: Received 100000 elements from Rank 6 (First element: 60)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat adjacent_trace.csv
2,3,400000,0.000293
4,5,400000,0.000391
0,1,400000,0.000395
6,7,400000,0.000482
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 16 ./mpi_adjacent 100000
--------------------------------------------------------------------------
There are not enough slots available in the system to satisfy the 16
slots that were requested by the application:

  ./mpi_adjacent

Either request fewer procs for your application, or make more slots
available for use.

A "slot" is the PRRTE term for an allocatable unit where we can
launch a process.  The number of slots available are defined by the
environment in which PRRTE processes are run:

  1. Hostfile, via "slots=N" clauses (N defaults to number of
     processor cores if not provided)
  2. The --host command line parameter, via a ":N" suffix on the
     hostname (N defaults to 1 if not provided)
  3. Resource manager (e.g., SLURM, PBS/Torque, LSF, etc.)
  4. If none of a hostfile, the --host command line parameter, or an
     RM is present, PRRTE defaults to the number of processor cores

In all the above cases, if you want PRRTE to default to the number
of hardware threads instead of the number of processor cores, use the
--use-hwthread-cpus option.

Alternatively, you can use the --map-by :OVERSUBSCRIBE option to ignore the
number of available slots when deciding the number of processes to
launch.
--------------------------------------------------------------------------
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano scaling_results.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns %
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat scaling_results.csv
Processes,Latency
2,0.000155
4,0.000663
8,0.000482
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano plot_scaling.py
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % python3 plot_scaling.py

=== SCALING DATA ===
   Processes   Latency
0          2  0.000155
1          4  0.000663
2          8  0.000482

Graph saved as scaling_graph.png
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cd ~/Downloads/SST_Stimulator/adjacent_experiment
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % ls
adjacent_communication.png      adjacent_latency.png            adjacent_stats.csv              Scaling
adjacent_graph.png              adjacent_sst.py                 plot_stats.py                   Untitled.pdf
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % rm -f adjacent_stats.csv
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % sst adjacent_sst.py
Simulation is complete, simulated time: 1.5235 us
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % ls
adjacent_communication.png      adjacent_latency.png            adjacent_stats.csv              Scaling
adjacent_graph.png              adjacent_sst.py                 plot_stats.py                   Untitled.pdf
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % head -40 adjacent_stats.csv
ComponentName,StatisticName,StatisticSubId,StatisticType,SimTime,Rank,Sum.u64,SumSQ.u64,Count.u64,Min.u64,Max.u64
cpu,pendCycle,,Accumulator,1523500,0,16865,156659,3047,0,16
cpu,reads,,Accumulator,1523500,0,590,590,590,1,1
cpu,writes,,Accumulator,1523500,0,410,410,410,1,1
l1cache,stateEvent_GetS_I,,Accumulator,1523500,0,10,10,10,1,1
l1cache,stateEvent_GetS_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetS_M,,Accumulator,1523500,0,569,569,569,1,1
l1cache,stateEvent_GetX_I,,Accumulator,1523500,0,6,6,6,1,1
l1cache,stateEvent_GetX_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetX_M,,Accumulator,1523500,0,394,394,394,1,1
l1cache,stateEvent_GetSX_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetSX_S,,Accumulator,1523500,0,0,0,0,0,0
```

```
l1cache,stateEvent_GetSX_M,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetSResp_IS,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetXResp_IS,,Accumulator,1523500,0,10,10,10,1,1
l1cache,stateEvent_GetXResp_IM,,Accumulator,1523500,0,6,6,6,1,1
l1cache,stateEvent_GetXResp_SM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_IS,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_IM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_SM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_SB,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_IB,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInvX_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInvX_M,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInvX_IS,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInvX_IM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInvX_SB,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInvX_IB,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_IS,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_IM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_SM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_IB,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Fetch_SB,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInv_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInv_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_FetchInv_M,,Accumulator,1523500,0,0,0,0,0,0
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % python3 plot_stats.py

=== AVAILABLE STATISTICS ===
<StringArray>
[                          'pendCycle',                                        'reads',
                              'writes',                           'stateEvent_GetS_I',
                   'stateEvent_GetS_S',                           'stateEvent_GetS_M',
                   'stateEvent_GetX_I',                           'stateEvent_GetX_S',
                   'stateEvent_GetX_M',                          'stateEvent_GetSX_I',
 ...
              'requests_received_PutM',                        'outstanding_requests',
                        'latency_GetS',                                'latency_GetSX',
                        'latency_GetX',                                'latency_Write',
                        'latency_PutM',                            'cycles_with_issue',
 'cycles_attempted_issue_but_rejected',                                 'total_cycles']
Length: 176, dtype: str

=== EXTRACTED METRICS ===
Reads: 590
Writes: 410
Cache_GetS: 569
Cache_GetX: 394
PendingCycles: 16865

Graph saved as adjacent_graph.png
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % rm -f adjacent_stats.csv
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % sst adjacent_sst.py
Simulation is complete, simulated time: 6.8695 us
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % python3 plot_stats.py

=== AVAILABLE STATISTICS ===
<StringArray>
[                          'pendCycle',                                        'reads',
                              'writes',                           'stateEvent_GetS_I',
                   'stateEvent_GetS_S',                           'stateEvent_GetS_M',
                   'stateEvent_GetX_I',                           'stateEvent_GetX_S',
                   'stateEvent_GetX_M',                          'stateEvent_GetSX_I',
 ...
              'requests_received_PutM',                        'outstanding_requests',
                        'latency_GetS',                                'latency_GetSX',
                        'latency_GetX',                                'latency_Write',
                        'latency_PutM',                            'cycles_with_issue',
 'cycles_attempted_issue_but_rejected',                                 'total_cycles']
Length: 176, dtype: str

=== EXTRACTED METRICS ===
Reads: 2970
Writes: 2030
Cache_GetS: 2949
Cache_GetX: 2014
PendingCycles: 57847

Graph saved as adjacent_graph.png
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % rm -f adjacent_stats.csv
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % sst adjacent_sst.py
Simulation is complete, simulated time: 13.3575 us
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % python3 plot_stats.py

=== AVAILABLE STATISTICS ===
<StringArray>
[                          'pendCycle',                                        'reads',
                              'writes',                           'stateEvent_GetS_I',
                   'stateEvent_GetS_S',                           'stateEvent_GetS_M',
                   'stateEvent_GetX_I',                           'stateEvent_GetX_S',
                   'stateEvent_GetX_M',                          'stateEvent_GetSX_I',
 ...
              'requests_received_PutM',                        'outstanding_requests',
                        'latency_GetS',                                'latency_GetSX',
                        'latency_GetX',                                'latency_Write',
                        'latency_PutM',                            'cycles_with_issue',
 'cycles_attempted_issue_but_rejected',                                 'total_cycles']
Length: 176, dtype: str

=== EXTRACTED METRICS ===
Reads: 6012
```

```
Writes: 3988
Cache_GetS: 5991
Cache_GetX: 3972
PendingCycles: 109025

Graph saved as adjacent_graph.png
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % rm -f adjacent_stats.csv
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % sst adjacent_sst.py
Simulation is complete, simulated time: 66.839 us
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % python3 plot_stats.py

=== AVAILABLE STATISTICS ===
<StringArray>
[                          'pendCycle',                                   'reads',
                              'writes',                     'stateEvent_GetS_I',
                   'stateEvent_GetS_S',                     'stateEvent_GetS_M',
                   'stateEvent_GetX_I',                     'stateEvent_GetX_S',
                   'stateEvent_GetX_M',                    'stateEvent_GetSX_I',
 ...
              'requests_received_PutM',                  'outstanding_requests',
                        'latency_GetS',                          'latency_GetSX',
                        'latency_GetX',                          'latency_Write',
                        'latency_PutM',                     'cycles_with_issue',
 'cycles_attempted_issue_but_rejected',                          'total_cycles']
Length: 176, dtype: str

=== EXTRACTED METRICS ===
Reads: 30014
Writes: 19986
Cache_GetS: 29993
Cache_GetX: 19970
PendingCycles: 519050

Graph saved as adjacent_graph.png
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano scaling_results.csv
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano plot_scaling.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % python3 plot_scaling.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % open overall_scaling_graph.png
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment %
```